Two components of the finite-time memory fluctuation
A conserved four-field correction and a mixed spectral check

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Abstract

The four-field term in the Gaussian Dyson expansion of the ordinary memory force has a strong leading energy regression. Independent auxiliary copies, three-color chord moments and the standard q-Gaussian conditional expectation prove this in annealed L^2, uniformly on compact real-time intervals. Exact energy pinching therefore removes its collective contribution. The actual memory field reduces to two bounded components: a charged-transfer defect and a neutral covariance-filter residual. We identify the first two neutral Taylor coefficients and their mixed spectral measures with the charged coefficients. A nonzero second-order cross is a necessary calibration for the still-unknown joint finite-time kernel.

1 Model, state and previously established input

Use standard CAR, $\hbar = 1$, string real time, and the overlapping complex p-body Gaussian model

$$H_N = \sum_{I,J} J_{IJ} c_I^\dagger c_J, \qquad \mathbb{E} J_{IJ} J_{KL} = w_N \delta_{IL} \delta_{JK}, \tag{1}$$

$$w_N = 4^p \frac{J_s^2}{\lambda_N} \left(\frac{p!}{N^p} \right)^2, \qquad \lambda_N = p^2/N. \tag{2}$$

All charges share the same original J. Take even N, fixed $J_s > 0$, and first $\lambda_N \to \lambda > 0$. The state is the normalized full $\beta = \mu = 0$ annealed trace, with $b_m = 2^{-N}\binom{N}{m}$. All L^r norms include this trace and the disorder average. At charge m write $h = H_m$ and

$$C_m = w_N \binom{m}{p} \binom{N-m+p}{p}, \quad M = \sup_m C_m \le (J_s^2/\lambda_N) e^{2p/N}. \tag{3}$$

Let P_e be the exact spectral projections of h and $\mathsf{P}_h X = \sum_e P_e X P_e$, including whole degenerate blocks. The preceding stages [4, 5] prove uniform second moments and $\sup_{|s|\le S} \|\mathsf{F}(s) - X_0(s)\|_2 \to 0$ for the actual memory field and its ordinary-propagation term. Here we further reduce X_0, keeping all common-disorder cross terms.

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2 Exact components and the new reduction

For $m > 0$ define

$$V\psi = m^{-1/2} \sum_i |i\rangle c_i \psi, \qquad K = 1_N \otimes H_{m-1}, \qquad Q = 1 - VV^\dagger, \tag{4}$$

$$R = QKV, \qquad D = QKQ, \qquad \alpha_m = (1 - p/m)_+, \tag{5}$$

$$\Lambda_m(s) = R^\dagger e^{isK} R, \qquad M_m(s) = R^\dagger e^{isD} R, \qquad \epsilon_m = 1 - \alpha_m^2. \tag{6}$$

The fields are

$$X_{0,m} = (1 - \mathsf{P}_h) m e^{-ihs/2} \Lambda_m(s) e^{-ihs/2}, \qquad \mathsf{F}_m = (1 - \mathsf{P}_h) m e^{-ihs/2} M_m(s) e^{-ihs/2}. \tag{7}$$

Inactive sectors $m < p$ have zero Hamiltonian and force. At $m = 0$ all collective fields are set to zero. For an independent copy u of h, let $\Phi_m(X) = \mathbb{E}_u[uXu]$ and

$$\Lambda_{\mathrm{OU},m}(s) = \epsilon_m \mathbb{E}_u[u e^{is(\alpha_m h + \sqrt{\epsilon_m} u)} u]. \tag{8}$$

The same u occurs in its three positions. Define, exactly at finite N,

$$\mathsf{A}_m(s) = (1 - \mathsf{P}_h) m e^{-ihs/2} [\Lambda_m(s) - \Lambda_{\mathrm{OU},m}(s)] e^{-ihs/2}, \tag{9}$$

$$\mathsf{B}_m(s) = (1 - \mathsf{P}_h) m \epsilon_m e^{-ihs/2} \Phi_m(e^{i\alpha_m hs}) e^{-ihs/2}. \tag{10}$$

Both use the original h; auxiliary averaging does not replace the physical charged force by an independent probe.

The prior even Dyson expansion is

$$\Lambda_{\mathrm{OU}}(s) = \epsilon \Phi(e^{i\alpha hs}) - \epsilon^2 \mathcal{I}_s^{(\alpha)}(h) + \mathcal{E}_s, \tag{11}$$

$$\mathcal{I}_s^{(\alpha)}(h) = s^2 \int_{0 \le y \le x \le 1} \mathbb{E}_u[u U_{s(1-x)} u U_{s(x-y)} u U_{sy} u] \, dx \, dy, \tag{12}$$

$$\|\mathcal{I}_s^{(\alpha)}\|_\infty \le \tfrac{3}{2} C_m^2 s^2, \tag{13}$$

$$\|\mathcal{E}_s\|_2 \le \epsilon^3 |s|^4 \sqrt{10395}\, C_m^3 / 24. \tag{14}$$

Here $U_t = e^{i\alpha ht}$. All three fourth-moment Wick pairings occur in $\mathcal{I}$. We prove

$$\boxed{\sup_{|s| \le S} \|\mathsf{F}(s) - \mathsf{A}(s) - \mathsf{B}(s)\|_2 \to 0.} \tag{15}$$

Each of A, B is uniformly L^2 bounded by the weighted-transfer and neutral-filter bounds of [4]. Thus their full four-term covariance sum replaces the actual covariance, uniformly for compact memory times and all real external times.

3 Why four auxiliary fields have strong energy regression

Let h, u, v now denote three orthogonal q-Gaussian variables with variance $C = J_s^2/\lambda$ and $q = e^{-4\lambda}$. Write E_h and E_{hv} for their trace-preserving conditional expectations. The standard second-quantization construction [1] gives

$$E_{hv}W(h,u) = E_hW(h,u), \qquad W(h,u) \in W^*(h,u). \tag{16}$$

Indeed, the one-particle projection onto $\mathrm{span}(h,v)$ annihilates every u tensor and retains the h tensors. Applied factorwise to Wick tensors it is the same projection as onto h. For two polynomials W_1, W_2 this proves

$$\tau_q[W_1(h,u)^\dagger W_2(h,v)] = \tau_q[(E_hW_1)^\dagger(E_hW_2)]. \tag{17}$$

This identity concerns two whole blocks. It does not assert classical independence of arbitrary interleaved colored words.

For the finite model let

$$W_N(h,u) = uf_1(h)uf_2(h)uf_3(h)u, \qquad T_N(h) = \mathbb{E}_u W_N(h,u), \tag{18}$$

where the f_i are fixed polynomials. Independence of auxiliary couplings u, v gives the exact replica identity

$$\mathbb{E}_h\tau T_N^\dagger T_N = \mathbb{E}_{h,u,v}\tau W_N(h,u)^\dagger W_N(h,v). \tag{19}$$

The three-color joint moments converge to those of h, u, v above. To see the color extension, perform Gaussian Wick expansion first: pairings are allowed only between the same color, but every allowed chord has the same independent p-set sums and CAR trace as before. The fixed-word overlap rule [2, 3] gives q per crossing, with the stated variance normalization. No extra covariance is introduced by color; finitely many pairings suffice at fixed degree. This argument holds for the full normalized trace. The established central-window trace bridge also gives its fixed-sector version uniformly for $m = N/2 + xp$ with bounded x [6].

Let $t(x)$ be the ordinary polynomial defined by $t(h) = E_hW(h,u)$. Mixed moments with $t(H_N)$ and (17) give

$$\boxed{\|T_N(H_N) - t(H_N)\|_2 \longrightarrow 0.} \tag{20}$$

It follows by expanding the squared norm: its three limiting terms are $\|t\|_2^2$, $-2\|t\|_2^2$, and $\|t\|_2^2$. An annealed mean alone would not establish this cancellation.

4 Full real time, charge weights and exact pinching

First set $\alpha = 1$ and define

$$T_{abc,N}(h) = \mathbb{E}_u[uh^a uh^b uh^c u], \qquad \mathcal{I}_s^{(1)} = \sum_{n\geq 0} \frac{s^2(is)^n}{(n+2)!} \sum_{a+b+c=n} T_{abc,N}(h). \tag{21}$$

Gaussian moments and Schatten Hölder with $n+4$ factors give

$$\|T_{abc,N}\|_2 \leq \sqrt{(2n+7)!!}\, M^{(n+4)/2}. \tag{22}$$

There are $\binom{n+2}{2}$ triples. The degree-n norm is therefore bounded, for $|s| \leq S$, by

$$\frac{S^{n+2}}{2n!}\sqrt{(2n+7)!!}\, M^{(n+4)/2}. \tag{23}$$

The sum converges for every fixed S, uniformly in N. Thus fixed-degree joint moments are sufficient: first truncate, then take $N \to \infty$, and finally remove the degree cutoff.

Define $\iota_s(h)$ by replacing the conditional average in $\mathcal{I}_s^{(1)}$ with E_h in the q-Gaussian algebra. BKS's C^* version [1] makes it a continuous function on $[-E_*, E_*]$, where $E_* = 2\sqrt{C/(1-q)}$. It obeys

$$\|\iota_s\|_\infty \leq \tfrac{1}{2}s^2 E_*^4. \tag{24}$$

Extend it to all real energies by $\widetilde{\iota}_s(x) = \iota_s(\chi(x))$, where χ clips to this interval. For each fixed polynomial truncation, Hamiltonian moment convergence and uniform integrability compare its evaluation at H_N with this bounded extension. The polynomial limit (20), the tail (23) and a finite time mesh consequently prove

$$\sup_{|s|\leq S} \|\mathcal{I}_s^{(1)}(H_N) - \widetilde{\iota}_s(H_N)\|_2 \to 0. \tag{25}$$

Time uniformity also follows directly from the normally convergent series; the bounded target is uniformly Lipschitz on compact s.

On $N/4 \leq m \leq 3N/4$, $1 - \alpha_m \leq 4p/N$. Telescoping the three unitaries, using their integral derivatives and Schatten-10 Hölder, bounds

$$\|\mathcal{I}_s^{(\alpha_m)} - \mathcal{I}_s^{(1)}\|_{2,m} \leq \tfrac{1}{2}|1-\alpha_m|\,|s|^3\sqrt{9!!}\, M^{5/2}. \tag{26}$$

Here $m\epsilon_m^2 \leq 16\lambda_N$. Outside that broad central interval, $m\epsilon_m^2 \leq N$ and the uniform operator bound on $\mathcal{I}$ combines with the binomial weight $2e^{-N/8}$. Since the bounded comparison commutes with h, (25) proves

$$\boxed{\sup_{|s|\leq S} \|(1-\mathsf{P}_h)m\epsilon_m^2 e^{-ihs/2}\mathcal{I}_s^{(\alpha_m)}e^{-ihs/2}\|_2 \to 0.} \tag{27}$$

Likewise $m\mathcal{E}_s \to 0$: centrally $m\epsilon_m^3 \leq 128\lambda_N^2/p$, and the exterior tail absorbs N^2. Together with the previous return theorem this proves (15).

5 An explicit expression for the conserved background

Let W_n be monic variance-C q-Hermite polynomials,

$$xW_n = W_{n+1} + C[n]_q W_{n-1}, \qquad [n]_q = (1-q^n)/(1-q). \tag{28}$$

Take $v = \alpha h + \beta u$, $\alpha^2 + \beta^2 = 1$ and $\beta^2 = \epsilon$. The left/right Wick tensor recursions of [1], followed by projection to pure h tensors, give

$$E_h[uW_n(v)u] = Cq^n\alpha^n W_n(h) + \epsilon C^2[n]_q[n-1]_q\alpha^{n-2}W_{n-2}(h). \tag{29}$$

The second term is zero for $n < 2$. Only two structures survive: the external u's pair with each other, or each contracts with an internal v. This keeps the same-copy effect in full.

Set $P_r W_n = r^n W_n$, $D_q W_n = [n]_q W_{n-1}$ and $\mathcal{N}_q W_n = nW_n$. Then, for $e_s(x) = e^{isx}$,

$$\Lambda^q_{\mathrm{OU}}(s) = \epsilon C P_{q\alpha} e_s(h) + \epsilon^2 C^2 P_\alpha D_q^2 e_s(h). \tag{30}$$

This extends from polynomials since the q-Gaussian fields are bounded and D_q^2 is bounded on their L^2 space. Its norm is at most $1/[C(1-q)]$ in the normalized Hermite basis.

Expanding along $\alpha = \sqrt{1-\epsilon}$ and comparing with the frozen-αh Dyson expansion gives

$$\boxed{\iota_s = \frac{C}{2} P_q[(\mathcal{N}_q - x\partial_x)e_s] - C^2 D_q^2 e_s.} \tag{31}$$

The $x\partial_x$ term comes from $e_s(\alpha h)$ and cannot be omitted. $\mathcal{N}_q$ is the energy-Hermite number operator, not the coupling-Gaussian chaos number at finite N. Its action on e_s is justified by the bounded Jacobi operator's factorial coefficient tail.

For a direct polynomial implementation write $x^n = \sum_j d_{n,j} W_j(x)$, with

$$d_{0,0} = 1, \qquad d_{n+1,j} = d_{n,j-1} + C[j+1]_q d_{n,j+1}. \tag{32}$$

If $S_k = \sum_{a+b+c=k} E_h[uh^a uh^b uh^c u]$, then

$$[S_k]_{W_j} = \tfrac{C}{2}(k+2-j)q^j d_{k+2,j} + C^2[j+2]_q[j+1]_q d_{k+2,j+2}. \tag{33}$$

In particular,

$$\iota_0 = \iota_0' = 0, \quad \iota_0'' = (2+q)C^2 I, \quad \iota_0''' = iC^2(1+4q+3q^2+q^3)h. \tag{34}$$

These are the leading conserved coefficients. Finite-N $\mathcal{I}_s$ need not commute with h; its dynamic part vanishes only in the specified double-scaling limit.

6 The first two neutral fluctuation coefficients

Write $\mathsf{B}(s) = isB_1 - s^2B_2/2 + O(s^3)$ at finite N and $\mathcal{J}_h X = \{h, X\}/2$. Direct differentiation gives

$$B_1 = m\epsilon\alpha(1 - \mathsf{P}_h)\Phi(h), \tag{35}$$

$$B_2 = m\epsilon(1 - \mathsf{P}_h)[\alpha^2\Phi(h^2) - \alpha\{h, \Phi(h)\}]. \tag{36}$$

Use $\kappa_r = C_m\mu(r)$ on physical ranks, with the positive-part extension when $2p$ lies outside them:

$$\mu(r) = \prod_{j=0}^{p-1} \left(1 - \frac{r}{m-j}\right)_+ \left(1 - \frac{r}{N-m+p-j}\right)_+ . \tag{37}$$

For active $m \geq p$, define

$$Z = h^2 - C_m I, \quad h = \sum_a h_a, \quad \mathbb{E}\,\mathrm{tr}_m\, h_a^2 = C_m\omega_a, \quad \bar{\kappa} = \sum_a \omega_a\kappa_a, \quad B_Z = \Phi(h) - \bar{\kappa}h. \tag{38}$$

The same coefficients can be written exactly as

$$B_1 = m\epsilon\alpha(1 - \mathsf{P}_h)B_Z, \tag{39}$$

$$B_2 = m\epsilon(1 - \mathsf{P}_h)[\alpha^2(\Phi - \kappa_{2p})Z - 2\alpha\mathcal{J}_h B_Z]. \tag{40}$$

Before pinching the difference of the two $B_2/(m\epsilon)$ expressions is the explicitly conserved polynomial

$$(\alpha^2\kappa_{2p} - 2\alpha\bar{\kappa})h^2 + \alpha^2 C_m(C_m - \kappa_{2p})I. \tag{41}$$

In a central compact window $m = N/2 + xp$, the exact product for μ gives, for fixed k, d,

$$p[\mu(kp - d) - \mu(kp)] \to 4\lambda q^k d, \qquad 0 \leq p[\mu(kp - d) - \mu(kp)] \leq 12\lambda_N d. \tag{42}$$

For example, differentiate the logarithm of its two products: the reciprocal sum is $4p/N + O(p^2/N^2)$ at fixed deficit, while $\mu(kp) \to q^k$. Input factorial tails and the prior sixth output-deficit moment [6, 7] make this an L^2 multiplier limit, not merely a pointwise rank estimate. The output squared tail is bounded by a constant times R^{-4} after a cutoff at deficit R.

Let $D_2 Z = \sum_r (2p - r)P_r Z$, $D_{\mathrm{in}} = \sum_a (p - a)h_a$, $\bar{u} = \sum_a (p - a)\omega_a$, and $L_{\mathrm{in}} = \mathsf{W}_2\{h, D_{\mathrm{in}} - \bar{u}h\}$. The exact input/output decomposition and the prior strong bridges are

$$D_2 Z = S_N + L_{\mathrm{in}} + 2\bar{u}Z, \qquad S_N = \sum_{a,b,r} (a + b - r)P_r \mathsf{W}_2(h_a h_b), \tag{43}$$

$$pB_Z = F/2 + o_{L^2}(1), \qquad \mathcal{J}_h F = 4\lambda C q L_{\mathrm{in}} + o_{L^2}(1), \tag{44}$$

$$S_N = (O_- + O_+)/2 + o_{L^2}(1). \tag{45}$$

Here $O_\pm = \mathsf{W}_2 \sum_i [H, c_i]^\dagger [H, c_i]$ and its reverse ordering, respectively; F is the previously defined linear rank-filtered field. With $Y_\pm = O_\pm - cZ$, $c = 2\lambda(1 - q)/(1 + q)$, substitution into (40) gives

$$\boxed{B_1 \to F, \qquad B_2 \to d(Y_- + Y_+) + b\mathcal{J}_h F, \quad d = 4\lambda C q^2, \quad b = 2(q - 1).} \tag{46}$$

Exact pinching can be removed from the limit because these prior Mehler spectral measures have no mass on the energy diagonal, including energy-polynomial descendants with their controlled higher moments. Multiplication of $pB_Z - F/2$ by $\mathcal{J}_h$ is justified: for an invariant Hermitian linear Gaussian error X, $\|\mathcal{J}_h X\|_2^2 \leq 3C_m\|X\|_2^2$ by its scalar variance and Gaussian fourth moment; independence of X and h is not needed.

7 A nonzero mixed spectral calibration

Use the established two-energy measures μ_r with Hamiltonian variance C and chord parameter q, normalized to mass one. Put $e = (E + E')/2$. The needed prior data [6, 8] are

$$d\nu_{(Y_-,Y_+)} = \begin{pmatrix} w & \eta \\ \eta & w \end{pmatrix} d\mu_{q^2}, \qquad d\nu_F = V_F d\mu_q, \tag{47}$$

$$w = 4C^2\lambda + \frac{16C^2\lambda^2 q}{1+q}, \qquad \eta = -4C^2\lambda q + \frac{16C^2\lambda^2 q}{1+q}, \tag{48}$$

$$V_F = 64 J_s^6 q^2, \qquad \nu_{F,Y_\pm} = 0. \tag{49}$$

The first two charged coefficients $\mathsf{A}(s) = A_0 + isA_1 + O(s^2)$ have the previously identified limits

$$A_0 \to Y_-, \qquad A_1 \to L - F = R + a\mathcal{J}_h Y_-, \qquad a = -\frac{(1-q)^2}{1+q^2}, \tag{50}$$

where $\nu_{R,F} = \nu_{R,Y_\pm} = 0$. Indeed $\mathsf{B}(0) = 0$, the four-field term starts at s^2, and the actual/ordinary memory fields have identical zeroth and first derivatives at zero.

Equation (46) yields

$$d\nu_{B_1,B_2} = beV_F\, d\mu_q, \tag{51}$$

$$d\nu_{B_2} = 2d^2(w+\eta)d\mu_{q^2} + b^2e^2V_F d\mu_q, \tag{52}$$

$$d\nu_{A_0,B_2} = d(w+\eta)d\mu_{q^2}, \qquad d\nu_{A_1,B_2} = ade(w+\eta)d\mu_{q^2}, \tag{53}$$

$$\nu_{A_0,B_1} = \nu_{A_1,B_1} = 0. \tag{54}$$

In particular $w + \eta > 0$ for $0 < q < 1$, so the second-order mixed measure is nonzero. Its value at zero external time is $d(w + \eta) > 0$. This is a coefficient-level result: any joint time kernel admitting termwise matching must reproduce it. We have not exchanged differentiation with an unproved finite-time large-N limit.

The coefficient measures extend to the full trace without assuming an arbitrary extension of every named limiting field. For $K_j = 2 \cdot 10^6(\lambda_N j^2 + \lambda_N^2 j^4)$, the broad-central rank bounds and Gaussian fourth moments give directly

$$\|B_1\|_{2,m} \le 24\lambda_N C_m^{3/2}\sqrt{K_1}, \tag{55}$$

$$\|B_2\|_{2,m} \le 24\lambda_N C_m^2[\sqrt{2K_2} + 2\sqrt{3K_1}]. \tag{56}$$

The ingredients are

$$\|B_Z\|_2 \le C_m^{3/2}(12p/N)\sqrt{K_1}, \qquad \|D_2 Z\|_2 \le \sqrt{2}C_m\sqrt{K_2},$$
$$\|\mathcal{J}_h B_Z\|_2 \le \sqrt{3}C_m^2(12p/N)\sqrt{K_1}.$$

All active sectors have the coarser bounds

$$\|B_1\|_{2,m} \le 2pC_m^{3/2}, \qquad \|B_2\|_{2,m} \le 6\sqrt{3}pC_m^2. \tag{57}$$

Central annuli contribute $O(e^{-2\lambda_- L^2})$, and the far tail contributes $O(p^2 e^{-N/8})$. The prior full-trace bounds on Y, L, together with spectral Cauchy–Schwarz, handle the mixed terms with A_0, A_1. The central coefficient limits therefore give the displayed full-trace spectral measures.

8 Verification, scope and the next calculation

Three new scientific outputs accompany this stage. Exact rational q-Fock calculations are compared with an independent colored-pairing enumeration: 2520 conditional trace equalities, 7200 three-color replica equalities, 66 same-copy OU Wick identities, and 42 four-field Taylor comparisons. Parameters are $q = 1/4, 1/2, 3/4$, $C = 1, 3/2$, and $\alpha = 3/5, \beta = 4/5$. The tests include every triple $a + b + c \leq 6$ and all three four-field Wick contractions.

Complete CAR coordinate sets in four systems give 24 real-time integrals, with time quadrature orders 8 and 12 and both signs of time. The maximum refinement difference is below $2 \cdot 10^{-15}$. An interior $N = 5, p = 2, m = 3$ sector has a nonzero finite-N pinched residual, confirming that the asymptotic regression is not an exact conserved identity. Initial test systems with $p = 1$ or an edge sector failed to supply this witness; the failed assertion and source are retained. No scientific output was overwritten.

A third script checks 12 complete-coordinate covariance/coefficient identities without floating energy pinching. It retains 360 exact rational slope inequalities and 80/120-digit comparisons for $\lambda = 1/4, 1/2, 1$, $p = 16, 32, 64, 128$, $x = -2, 0, 2$, $k = 1, 2$, and deficits $0, 1, 2, 3, 6$. Relative finite-size errors are retained. These computations check finite identities and conventions; the uniform limits come from the proofs above. Quadrature refinement is not interval certification.

Remaining work. The finite-time field now has a two-component reduction with the full covariance $C_{AA} + C_{AB} + C_{BA} + C_{BB}$. Its joint limiting kernel has not been evaluated. The next calculation must retain the charged defect's reverse-source contribution and the neutral rank-filter fluctuation with the same original J, then resum their mixed marked words in both memory times. The second-order cross above is a concrete check on that construction. No bulk mass assignment or finite-G interaction law follows yet. If λ_N merely stays in a positive compact interval, (15) still holds by taking convergent subsequences; explicit q, C spectral formulas assume a chosen limit λ.

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